ZCAS UNIVERSITY

CIT2241 BUSINESS INFORMATION SYSTEMS

DEFFERED EXAMINATIONS

THURSDAY, 18th JUNE 2026

09:00-12:00 HOURS

TIME ALLOWED: THREE (03) HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS TO CANDIDATES:

a) **SECTION A:** This section is compulsory. You must answer **Question 1.**

b) **SECTION B:** Attempt **any three (03)** questions out of the **four (04)** questions provided.

c) Time allowed: **3 Hours plus five Minutes to read through the paper.**

d) This examination paper carries a total of **100 marks.**

e) Candidates **MUST NOT** turn this page **until** the Invigilator tells them to do so.

SECTION A: This section is compulsory. You must answer **Question 1.**

QUESTION ONE (40 MARKS)

CASE STUDY

Kitwe Central Hospital has operated a manual patient records system for over twenty years. Patient records are stored in paper files distributed across outpatient, inpatient, laboratory, pharmacy, maternity, and specialist departments. As the hospital continues to grow, the manual system has become increasingly inefficient.

Patient files are often misplaced, duplicated, or retrieved late, causing delays in diagnosis and treatment. Management struggles to generate accurate reports on disease patterns, medicine usage, staff performance, and resource allocation. The hospital has also experienced incidents of unauthorized access to sensitive patient records.

To improve service delivery, management plans to implement a computerized Electronic Health Records (EHR) system. However, concerns have been raised regarding implementation costs, staff resistance, cybersecurity threats, data migration errors, system downtime, and legal compliance requirements.

As a Business Information Systems Consultant, evaluate the proposed project.

REQUIRED:

a) Critically evaluate how failure in any TWO phases of the SDLC could result in project failure. Use examples from the case study to support your answer. (5 Marks)

b) The hospital records the number of patients attended per day as follows:

	A	B
1	**DAY**	**PATIENTS**
2	Monday	15
3	Tuesday	28
4	Wednesday	22
5	Thursday	18
6	Friday	35

Use **Excel formula** to:

i. Calculate the total number of patients attended. (2 Marks)
ii. Calculate the average number of patients attended per day. (2 Marks)
iii. Determine the highest number of patients attended in a day. (2 Marks)
iv. Display "Overloaded" if patients exceed 25 and "Normal" otherwise. (2 Marks)
v. Count the number of days where attendance exceeded 20 patients. (2 Marks)

c) Evaluate FIVE critical factors that management must consider before acquiring and implementing the EHR system. Justify each factor. (5 Marks)

d) Using the DIKW and RAKID models, demonstrate how raw patient data can be transformed into actionable healthcare decisions. Discuss the consequences of poor-quality data at each stage. (5 Marks)

e) Using Anthony's Triangle, design an information systems framework for the hospital by recommending suitable systems for the three management levels. Justify your recommendations. (5 Marks)

f) Analyze FIVE major risks associated with computerized healthcare systems and explain their potential impact on hospital operations. (5 Marks)

g) Design a multi-layered information security strategy for the hospital using the People, Process, and Technology framework. (5 Marks)

TOTAL: 40 MARKS

SECTION B
ANSWER ANY THREE (03) QUESTIONS

QUESTION TWO (20 MARKS)

CASE STUDY

EDUTECH College recently implemented a Student Information System to manage admissions, course registration, examinations, fees, and reporting.

After deployment, users reported frequent system errors, poor performance, security vulnerabilities, and low adoption rates. Lecturers complain that they were not adequately consulted during requirements gathering. Management repeatedly requested new features during development, resulting in budget overruns and project delays.

REQUIRED:

a) Determine which SDLC phases were most likely poorly executed and justify your answer using evidence from the case study. (5 Marks)

b) Critically analyze the consequences of poor requirements gathering during systems analysis. Recommend measures that could have improved requirements quality. (5 Marks)

c) Compare and contrast the Waterfall and Agile SDLC models. Based on the case study, recommend the most suitable methodology and justify your choice. (5 Marks)

d) Evaluate the risks of implementing a system without adequate testing, security controls, and user training. Propose mitigation strategies for each identified risk. (5 Marks)

Total (20 marks)

QUESTION THREE (20 MARKS)

a) Discuss **CIA Triad, Repudiation and Non-repudiation**, and explain how an organization can protect its information. (5 marks)

b) With examples, identify the main categorization of Security Threats and describe the structured and unstructured type of threats. (5 marks)

c) As a BIS expert, distinguish **Encapsulation** from **Encryption** in information systems or software development. (5 marks)

d) Define these terms; **"Evil Twin"**, **"Script Kiddies"**, **"Hackitivists"** **"DDOS attack"**, **"Phishing"**. (5 marks)

Total (20 marks)

QUESTION FOUR (20 MARKS)

CASE STUDY

A Kantemba grocery store owner in Lusaka plans to expand the business into five branches within the next three years. Currently, all sales, inventory records, supplier information, and financial transactions are managed manually.

The owner is considering investing in Business Information Systems to support future growth.

REQUIRED:

a) Critically evaluate whether the business should continue using manual systems or adopt computerized Business Information Systems. (5 Marks)

b) Analyze FIVE risks associated with digital transformation and propose suitable mitigation measures. (10 Marks)

c) Explain how Business Intelligence and Analytics can provide competitive advantage to the expanding business. (5 Marks)

Total (20 marks)

QUESTION FIVE (20 MARKS)

CASE STUDY

Masambililo College recently implemented a new Student Management System. Two weeks before final examinations, the college experiences a ransomware attack that encrypts examination records, student results, and financial data.

Lecturers are unable to access marks, students cannot register for examinations, and management fears reputational damage.

REQUIRED:

a) Using the CIA Triad, Repudiation, and Non-Repudiation principles, analyze the impact of the ransomware attack on the college. (5 Marks)

b) Critically evaluate the different categories of security threats and distinguish between structured and unstructured threats using relevant examples.
(5 Marks)

c) Distinguish between Encapsulation and Encryption. Explain why confusing the two concepts could lead to poor security decisions.
(5 Marks)

d) Design a comprehensive cybersecurity strategy for the college using the People, Process, and Technology framework. Justify each security control recommended.
(5 Marks)

Total (20 marks)

END OF EXAMINATIONS